

SOLUTIONS

1. C is correct. The nominal risk-free rate is approximated as the sum of the real risk-free interest rate and an inflation premium.
2. C is correct. US Treasury bonds are highly liquid, whereas the bonds of small issuers trade infrequently and the interest rate includes a liquidity premium. This liquidity premium reflects the relatively high costs (including the impact on price) of selling a position. As the two bond issues have the same 30-year maturity, the observed difference in yields would not be solely explained by maturity. Further, the inflation premium embedded in the yield of both bonds is likely to be similar given they are both US-based bonds with the same maturity.
3. A is correct. The geometric mean return for Fund Y is calculated as follows:

$$\bar{R}_G = [(1 + 0.195) \times (1 - 0.019) \times (1 + 0.197) \times (1 + 0.350) \times (1 + 0.057)]^{(1/5)} - 1$$

$$= 14.9\%.$$
4. A is correct. The harmonic mean is appropriate for determining the average price per unit as it gives equal weight to each data point and reduces the potential influence of outliers. It is calculated as follows:

$$\bar{X}_H = 4 / [(1/62.00) + (1/76.00) + (1/84.00) + (1/90.00)] = \text{EUR}76.48.$$
5. B is correct. The geometric mean compounds the periodic returns of every period, giving the investor a more accurate measure of the terminal value of an investment.
6. C is correct. The money-weighted rate of return considers both the timing and amounts of investments into the fund. To calculate the money-weighted rate of return, tabulate the annual returns and investment amounts to determine the cash flows.

Year	1	2	3
Balance from previous year	0	USD1,150	USD4,560
New investment	USD1,000	USD2,850	USD40,440
Net balance at the beginning of year	USD1,000	USD4,000	USD45,000
Investment return for the year	15%	14%	-4%
Investment gain (loss)	USD150	USD560	-USD1,800
Balance at the end of year	USD1,150	USD4,560	USD43,200

$$CF_0 = -\text{USD}1,000, CF_1 = -\text{USD}2,850, CF_2 = -\text{USD}40,440, CF_3 = +\text{USD}43,200.$$

$$CF_0 = -1,000$$

$$CF_1 = -2,850$$

$$CF_2 = -40,440$$

$$CF_3 = +43,200$$

$$\frac{CF_0}{(1 + IRR)^0} + \frac{CF_1}{(1 + IRR)^1} + \frac{CF_2}{(1 + IRR)^2} + \frac{CF_3}{(1 + IRR)^3}$$

$$= \frac{-1,000}{1} + \frac{-2,850}{(1 + IRR)^1} + \frac{-40,440}{(1 + IRR)^2} + \frac{43,200}{(1 + IRR)^3} = 0$$

Solving for *IRR* results in a value of *IRR* = -2.22 percent.